The Role of Temperature Gradients versus Static Stability on the Zonal Wind and Eddy Kinetic Energy Response to Thermal Perturbations

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ABSTRACT: The relative role of meridional temperature gradients and static stability on the extratropical response to thermal perturbations are explored in a dry Primitive Equation (PE) GCM. A Quasi-Geostrophic (QG) model is used to separate the relative roles because the static stability in a QG model is externally prescribed and can therefore be altered independently of the meridional temperature gradient. For most experiments, the changes in meridional temperature gradients make the largest contribution to the zonal-mean zonal wind (U) changes, although the static stability changes are also important. In most cases, the mechanism for the static stability response does not directly involve the eddies. Instead, the advection of the static stability perturbation by the zonal-mean vertical velocity accounts for almost all of the response to static stability. For increased static stability, advection by the Ferrel cell increases the meridional temperature gradient, which strengthens the jet via thermal wind. The stronger jet then shifts poleward, consistent with various theories in the literature. For Eddy Kinetic Energy (EKE), the direct effect of static stability makes a non-negligible contribution in most cases. In all cases, however, the meridional temperature gradient and advection by Ferrel cell effects are together more dominant.

1. Introduction

The response of the atmosphere to idealized thermal perturbations (e.g. Haigh et al. (2005); Williams (2006); Lorenz and DeWeaver (2007); Frierson et al. (2007); Frierson (2008); Butler et al. (2010); McGraw and Barnes (2016); Wu and Smith (2016); Mbengue and Schneider (2017)) is a good starting point to understanding the response to increased greenhouse gases in comprehensive climate models. In this paper we focus on the response of the extratropical zonal-mean zonal winds (U) and Eddy Kinetic Energy (EKE) in a Primitive Equation (PE) and a Quasi-Geostrophic (QG) model on a sphere. The goal is to understand and quantify the relative role of meridional temperature gradients versus static stability (i.e. vertical temperature gradients) on the response to thermal perturbations. In particular, in the QG model, the static stability is prescribed a priori, so the temperature gradients and static stability can be changed independently to understand the total response.

Previous studies have reached differing conclusions on the relative role of static stability versus meridional temperature gradients. For example, Frierson et al. (2006), Frierson (2008) and Lu et al. (2008) argue that the increased static stability due to global warming causes the poleward shift of the jet (U) and storm tracks (EKE): increased static stability causes the subtropical boundary between baroclinically stable/unstable to shift poleward thereby shifting the baroclinic wave generation and eddy momentum flux convergence poleward. Poleward shifts in the subtropical edge of the Hadley circulation have also been attributed to static stability via the same mechanism (Frierson et al. (2007); Kang and Lu (2012); Chemke and Polvani (2019)), however, the eddy driven jet and Hadley cell edge are not necessarily closely related for forced changes (Kang and Polvani (2011)). Other studies also believe static stability is important, but for different reasons. For example, Kidston et al. (2010) and Kidston et al. (2011) argue that increased static stability increases the horizontal length scale of the eddies which then shifts the jet poleward via interactions with the critical level on the poleward flank of the jet. Interestingly, all the static stability mechanisms above involve the effect of static stability directly on the eddies. The response of U then comes about via the eddy response. However, we find that the effect of static stability directly on the zonal mean is the larger effect (vertical advection of the vertical potential temperature gradients by the zonal-mean vertical velocity).

Other studies implicate meridional temperature gradients on the extratropical response. For example, Mbengue and Schneider (2017) find that the latitude of EKE maximum is primarily affected by surface temperature gradients. Harvey et al. (2014), Barnes and Screen (2015) and Harvey et al. (2015) discuss the response to thermal perturbations via their effect on meridional temperature gradients. Changes in meridional temperature gradients are also the essential component for dynamical mechanisms that invoke the paradigm that stronger jets shift poleward (Chen et al. (2007); Kidston and Vallis (2012); Lorenz (2014a); Lorenz (2014b)).

In this paper we perform simulations in the PE model with various thermal perturbations to the Held and Suarez (1994) forcing as in the following studies: Haigh et al. (2005), Williams (2006), Gerber and Vallis (2007), Lorenz and DeWeaver (2007), Butler et al. (2010) and McGraw and Barnes (2016). Next we perform analogous simulations in the QG model. The full response in the QG model is a combination of 1) perturbations to meridional temperature gradients from the Held and Suarez (1994) forcing and 2) the vertical static stability profile taken from the corresponding PE simulation. Next the two components are applied separately to quantify the relative contribution of meridional temperature gradients and static stability. The static stability response can be further decomposed into the contribution of vertical advection of the vertical potential temperature gradients by the zonal-mean vertical velocity and everything else.

The paper begins with a description of the PE and QG models. Next the mean state and response to thermal perturbations is given for the PE model. The the QG model is used to understand the relative contribution of the meridional temperature gradients and static stability in the PE simulations. Finally a suite of QG simulations with perturbed static stability at various vertical levels is analyzed to better understand the dynamics.

2. Methods

a. QG Model

Most QG models on a sphere assume a constant Coriolis parameter in both the geostrophic relationship and the stretching term in the QG Potential Vorticity (PV). Here, we instead use a QG model on a sphere based on the linear balance equation (Whitaker and Loughe (1998)):

$$\nabla^2\left(\phi_j\right) = \nabla \cdot \left(f \nabla\left(\psi_j\right)\right), \tag{1}$$

where ψ is the streamfunction, f is the true Coriolis parameter, ϕ is the geopotential, and j is an index over pressure level. To be consistent with the literature of this model, j increases from the surface to the top of the atmosphere. The model is also formulated under the assumption of equally spaced pressure level (Δp), so that $p_j = 1000 + (j - \frac{1}{2})\Delta p$ mb. Lorenz (1960) developed the original model, but the vertical differencing is typically changed to a Charney-Phillips grid, and that is the version used here (Whitaker and Loughe (1998)). The vorticity equation is:

$$\frac{\partial \nabla^2 \psi_j}{\partial t} + J\left(\psi_j, f + \nabla^2 \psi_j\right) + \nabla \cdot \left(f \nabla \chi_j\right) = -k_M \nabla^2 \psi_j - \nu \left|\nabla^{2n}\right| \nabla^2 \psi_j, \tag{2}$$

where $J(\ldots)$ is the advection term written with the Jacobian (see below), χ is the velocity potential, k_M is the Rayleigh damping parameter, ν is the hyperdiffusivity parameter, and n is the order of the hyperdiffusion operator. The Jacobian is given by:

$$J(A, B) = \frac{1}{a^2} \frac{\partial A}{\partial \lambda} \frac{\partial B}{\partial \mu} - \frac{1}{a^2} \frac{\partial B}{\partial \lambda} \frac{\partial A}{\partial \mu}, \tag{3}$$

where a is the radius of the earth, λ is the longitude and μ is the sine of latitude. The vorticity equation (2) is the prognostic equation for the model, however, (2) includes the unknown variable χ. To close the system, one must consider the thermodynamic energy equation, which is given here in terms of the geopotential difference between levels:

$$\begin{aligned} \frac{\partial}{\partial t}\left(\phi_{j+1} - \phi_j\right) + \frac{1}{2} J\left(\psi_j + \psi_{j+1}, \phi_{j+1} - \phi_j\right) - \sigma_{j+1/2} \omega_{j+1/2} = \\ -k_T\left(\phi_{j+1} - \phi_j - \phi_E\right) - \nu \left|\nabla^{2n}\right| \left(\phi_{j+1} - \phi_j\right), \end{aligned} \tag{4}$$

where $\sigma_{j+1/2}$ is the QG static stability, $\omega_{j+1/2}$ is the vertical velocity, k_T is the thermal relaxation parameter and ϕ_E is the "radiative equilibrium" geopotential difference. All of these quantities are defined on half levels. The velocity potential, χ, is related to the vertical velocity by

$$\nabla^2 \chi_j = \frac{\omega_{j+1/2} - \omega_{j-1/2}}{\Delta p}, \tag{5}$$

and with boundary conditions $\omega_{1/2} = 0$ and $\omega_{N+1/2} = 0$, where N is the number of vorticity levels. To eliminate the unknown variable χ in (2), take $\nabla \cdot \left(f \nabla \left(\nabla^{-2}\right)\right)$ of the difference of adjacent levels

of (2) and ∇^2 of (4) and then use the balance equation, (1), to eliminate the time derivatives. This leads to a diagnostic "QG omega equation" for χ, which then is used to step the prognostic the vorticity equation, (2), forward in time. For details see Whitaker and Loughe (1998).

The QG static stability is given by:

$$\sigma_{j+1/2} = c_p \left(\left(\frac{p_j}{p_0} \right)^{\kappa} - \left(\frac{p_{j+1}}{p_0} \right)^{\kappa} \right) \frac{\Theta_{j+2} + \Theta_{j+1} - \Theta_j - \Theta_{j-1}}{4\Delta p}, \tag{6}$$

where c_p is the specific heat at constant pressure, $\kappa = 2/7$, and Θ is the horizontally uniform background potential temperature. The unusual Θ differencing in the last factor is the effective $\Delta\Theta/\Delta p$ stencil for a PE model with second order differencing in the vertical. This form is also less sensitive to noise in Θ and is approximately equal to $\frac{\Theta_{j+1}-\Theta_j}{\Delta p}$ for smooth profiles. At the two boundaries, one should set either $\Theta_{j+2} = \Theta_{j+1}$ or $\Theta_j = \Theta_{j-1}$, which is also based on the PE differencing. In this paper, the background Θ_j is taken from the pressure-level PE model, which is described below. The forcing in the QG model (k_m, k_T, ϕ_E) is from Held and Suarez (1994). The hyperdiffusion is 8th order with a time scale of 0.1 days for the smallest scale waves. Third-order Adams-Bashforth time differencing (Durran (1991)) is used. The resolution is T42 with 20 equally spaced pressure levels.

b. PE Model

Like the QG model, the PE model is also in pressure coordinates (Rivier et al. (2002)):

$$\begin{aligned} \frac{\partial \eta}{\partial t} &= -\frac{1}{a(1-\mu^2)} \frac{\partial A}{\partial \lambda} - \frac{1}{a} \frac{\partial B}{\partial \mu}, \\ \frac{\partial \delta}{\partial t} &= -\frac{1}{a(1-\mu^2)} \frac{\partial B}{\partial \lambda} - \frac{1}{a} \frac{\partial A}{\partial \mu} - \nabla^2 \phi - \nabla^2 E, \\ \frac{\partial \theta}{\partial t} &= -\frac{1}{a(1-\mu^2)} \frac{\partial C}{\partial \lambda} - \frac{1}{a} \frac{\partial C}{\partial \mu} - F, \end{aligned} \tag{7}$$

where η is the absolute vorticity, δ is the divergence, θ is the potential temperature and

$$
\begin{aligned}
A &= U\eta + V\delta + \frac{\partial V\omega}{\partial p}, \\
B &= V\eta - U\delta - \frac{\partial U\omega}{\partial p}, \\
C &= U\theta, \\
D &= V\theta, \\
E &= \frac{U^2 + V^2}{2(1-\mu^2)}, \\
F &= \frac{\partial \omega\theta'}{\partial p},
\end{aligned}
\tag{8}
$$

where U and V are $u\cos(\text{latitude})$ and $v\cos(\text{latitude})$, respectively, and θ' is the deviation from the reference, isothermal θ used for the semi-implicit method. The upper and lower boundary conditions are $\omega = 0$, which implies that the net divergence in a vertical column is zero. To force this contraint, we use the divergence difference, $\hat{\delta}_{j+1/2} = \delta_{j+1} - \delta_j$, at $n-1$ levels as a prognostic variable (where n is the number of vertical levels). The full divergence, which is required in (8), is calculated from the $\hat{\delta}$ definition and the contraint that the mass-weighted vertical integral of δ is zero. Detailes of the vertical differencing are found in Rivier et al. (2002). Like the QG model, the resolution is T42 with 20 equally spaced pressure levels. The model is forced with the diabatic heating and frictional damping of Held and Suarez (1994). An 8th order hyperdiffusion with a time scale of 0.1 days for the smallest scale waves is added to the three pronostic variables. The time differencing uses the semi-implicit AB3-AI2 method of Durran and Blossey (2012).

Despite the constant surface pressure assumption, the circulation of the pressure coordinate model appears to be unaffected. For example, we have compared the model to sigma coordinate models with two separate the vertical differencing schemes (Hoskins and Simmons (1975) and Simmons and Burridge (1981)). At low resolution (T42), the jet latitude and persistence of the sigma coordinate models are farther apart from each other than either is to the pressure coordinate model (i.e. the pressure coordinate model is the median model). At higher resolution, all models converge to the same circulation. We are also aware of the potential impact of constant surface pressure on the transient response of the zonal-mean zonal wind to forcing and heating (Haynes and Shepherd (1989)), however, the EOF patterns appear to be essentially unaffected.

TABLE 1. Perturbation experiments with Held and Suarez (1994) GCM.

Name	Description	Control Value	Perturbed Value
TSTRAT	decrease stratosphere temp.	T_{strat} = 200 K	T_{strat} = 195 K
$\Delta\theta_z$TROP	increase tropical static stability	$\Delta\theta_z$ = 10 K	$\Delta\theta_z$ = 15 K
ΔT_y	increase pole-to-eq. ΔT but keep tropopause height at 45° same	ΔT_y = 60 K, $T_{\max}$ = 315 K	ΔT_y = 75 K, $T_{\max}$ = 322.5 K
POLE	heating at low-levels over pole (polar amplification)	none	see text

c. Perturbation Experiments

We perform a set of perturbation experiments with localized static stability changes. For each experiment the σ perturbation is the form of a Gaussian function:

$$\sigma_{j+1/2} = c_p \left(\left(\frac{p_j}{p_0} \right)^{\kappa} - \left(\frac{p_{j+1}}{p_0} \right)^{\kappa} \right) \frac{A \exp\left(\left(\frac{p-p_c}{w} \right)^2 \right)}{4\Delta p}, \tag{9}$$

where the A is 2K, w is 100mb and p_c is the location of the perturbation. We perform ten simulations with different p_c ranging from 50mb to 950mb in increments of 100mb. Recall that for consistent energetics, the static stability of a QG model is constant in the horizontal so the above perturbations depend on pressure only.

3. Results

a. Mean State

The time-mean state in the PE and QG model are shown in Fig. 1 and 2, respectively. For the zonal-mean zonal wind (U), the mid-latitude jet is located around 42° in the PE model. In the QG model, the jet is shifted poleward (47°) and the "shoulder" of stronger subtropical U is more prominent. The near surface U is also stronger in the QG model. For eddy kinetic energy (EKE, panel b), the QG field is also further poleward, but the amplitude is slightly weaker. For the eddy momentum flux ($\overline{u'v'}$, panel c), the amplitude of the QG field is significantly stronger than for the PE model, which is consistent with the stronger surface U in the QG model. The weaker $\overline{u'v'}$ but stronger EKE of the PE model suggests that the PE jet is more like a waveguide than the QG jet (waveguide: waves are trapped in latitude due to reflection on the jet flanks). Also, consistent with the waveguide hypothesis is the fact that the vertical EP flux (panel d), or baroclinic wave source,

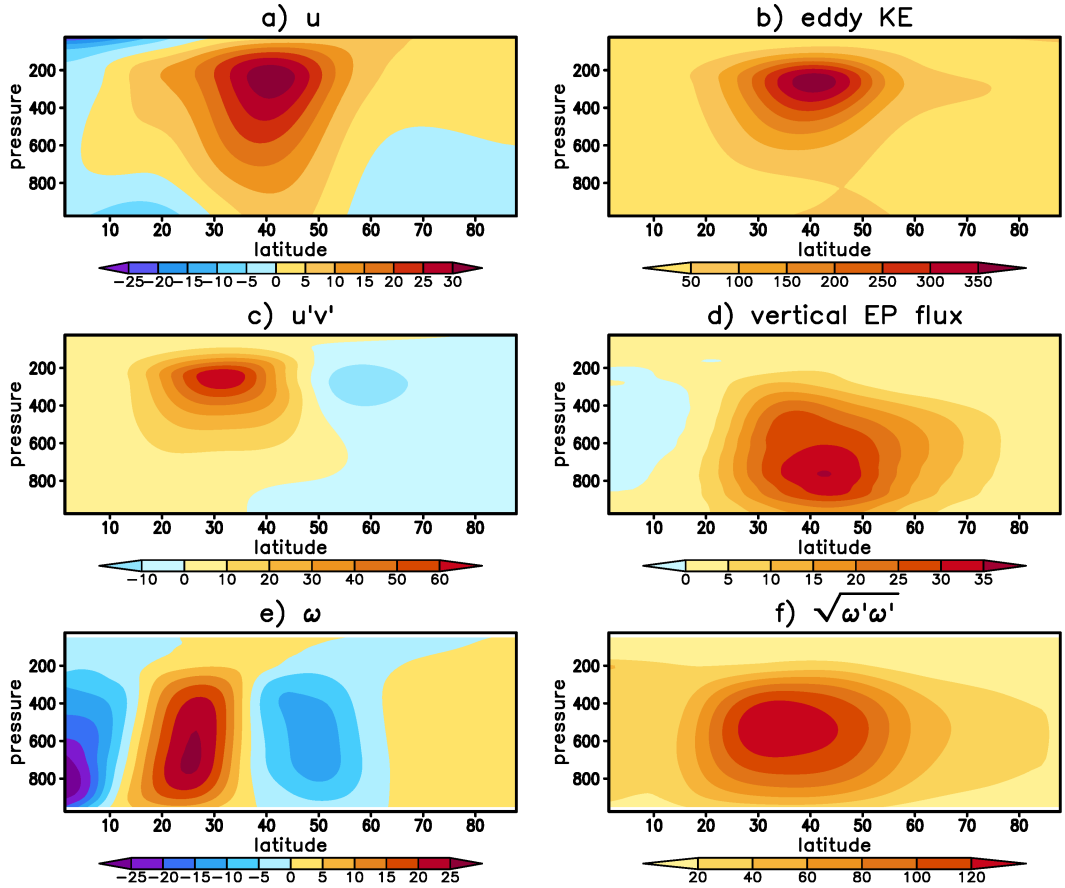

FIG. 1. Time-mean fields from PE GCM. a) zonal-mean zonal wind. b) zonal-mean eddy kinetic energy. c) zonal-mean eddy momentum flux. d) vertical component of the EP flux. e) zonal-mean pressure vertical velocity (ω). f) square root of the zonal-mean of the eddy ω variance.

is weaker in the PE model yet the EKE is stronger. So in other words, the waves that are created in the PE model linger in the mid-latitudes longer, whereas in the QG model they have a greater tendency to propagate to the subtropics where they dissipate at the critical line. We believe this waveguide issue is very interesting and it appears that there is more to the problem than simply the jet latitude, but we currently do not have a good understanding of this issue. Therefore, this is not discussed further in this paper.

Finally, we show vertical velocity ($= \omega$) statistics since this field is important for understanding the results below. In the tropics and subtropics, the zonal-mean ω circulation is stronger in the PE model (panel e). In the mid-latitudes, on the other hand, the ω circulation is about the same. For the eddy standard deviation (panel f), the QG model field is larger.

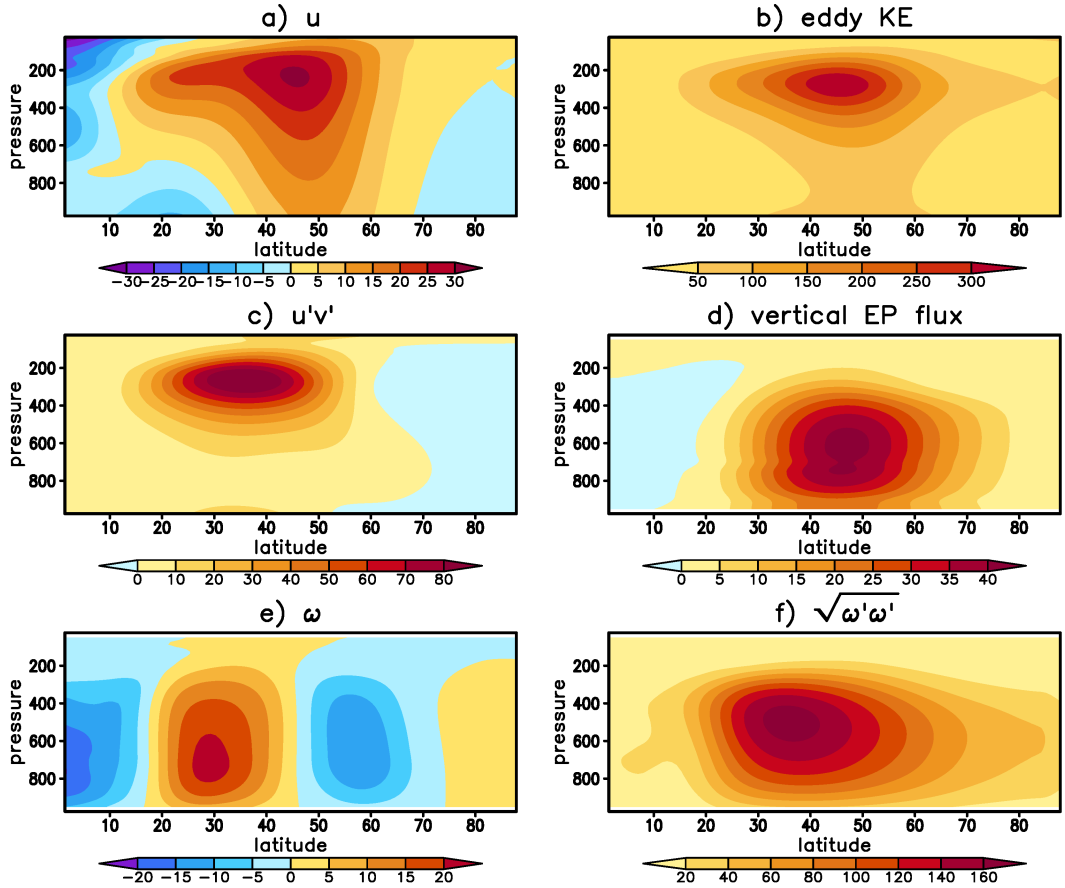

FIG. 2. Same as Fig. 1 but for the QG model.

b. Response to Thermal Forcings

The response of U and EKE in the PE model for the experiments in Table 1 are shown in Fig. 3. As shown in Haigh et al. (2005), Williams (2006) and Lorenz and DeWeaver (2007), decreasing the Held and Suarez (1994) stratosphere temperature raises the tropopause and causes a poleward shift the mid-latitude jet. The EKE also shifts poleward and upward in the TSTRAT simulation. For the $\Delta\theta_z$TROP experiment the jet also shifts poleward (Butler et al. (2010); McGraw and Barnes (2016)). The U changes look similar to TSTRAT except the U changes equatorward of 40° are larger in amplitude. For the EKE, the decreases in EKE on the equator flank of the jet are much more pronounced compared to TSTRAT and there is much less upward shift. For the ΔT_y experiment (e.g. Gerber and Vallis (2007)), the U changes on the poleward flank of the jet are more pronounced compared to the previous experiments and the EKE response is a monopole (Fig.

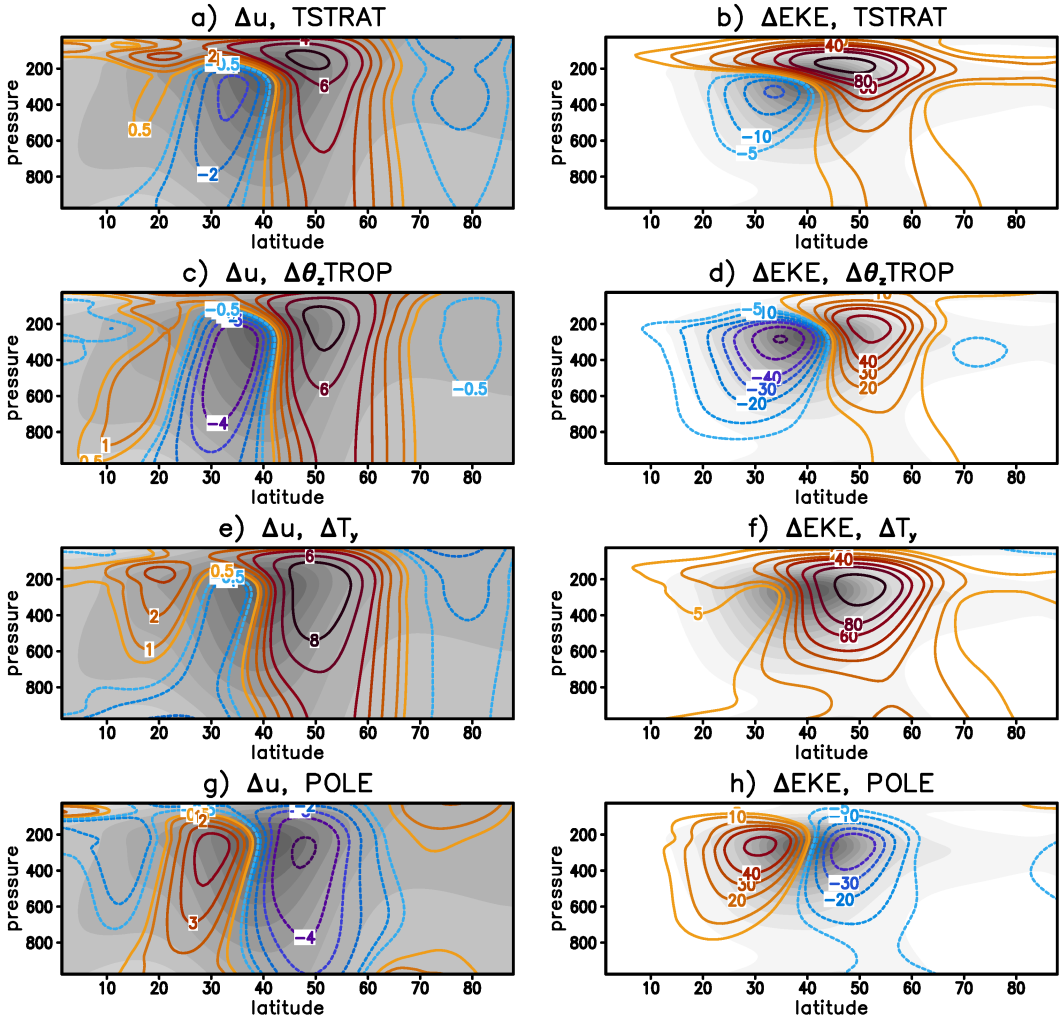

FIG. 3. PE results: a) Change in zonal-mean zonal wind for the TSTRAT experiment (see Table 1). b) Change in eddy kinetic energy for the TSTRAT experiment. c) Same as (a) but for the $\Delta\theta_z$TROP experiment. d) Same as (b) but for the $\Delta\theta_z$TROP experiment. e) Same as (a) but for the ΔT_y experiment. f) Same as (b) but for the ΔT_y experiment. g) Same as (a) but for the POLE experiment. h) Same as (b) but for the POLE experiment.

3f), unlike other simulations. Unlike the other experiments, polar amplification (POLE) causes an equatorward shift of U and EKE (Butler et al. (2010)).

c. Temperature Gradients versus Static Stability

In this subsection, we look at the change experiments (Table 1) in the QG model and we diagnose the relative role of meridional temperature gradients ($\partial\theta/\partial y$) and static stability ($d\Theta/dp$) for the U and EKE changes. Like any QG model, the static stability is prescribed, therefore to best represent

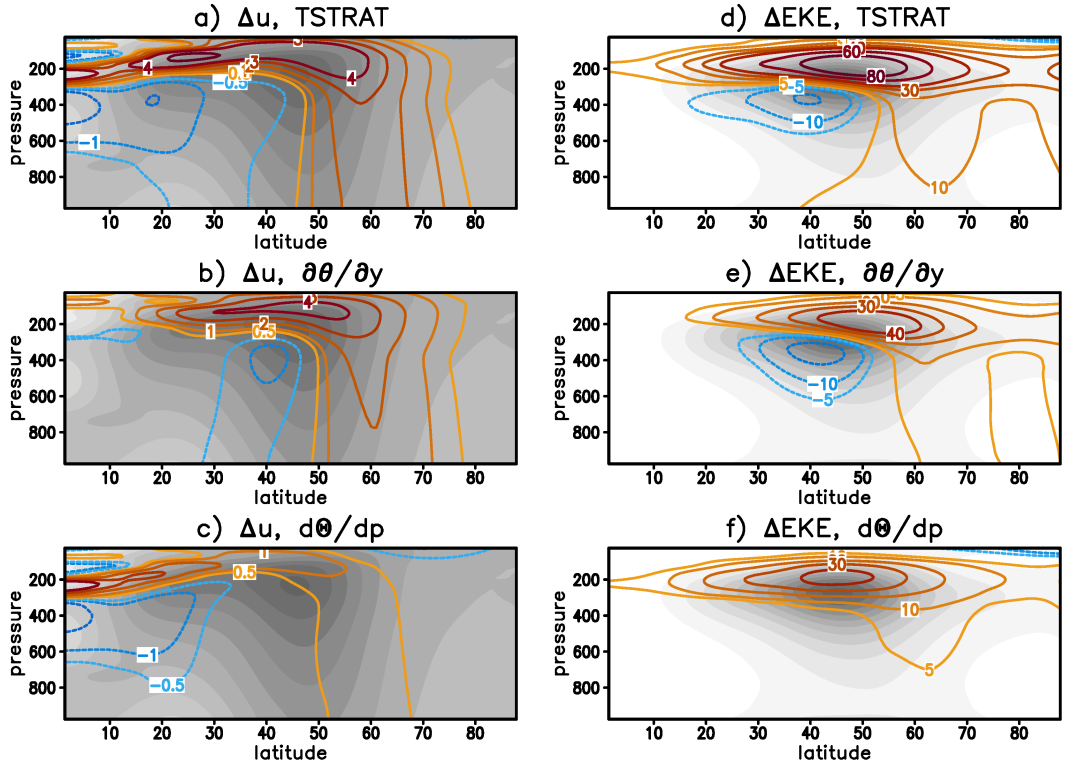

FIG. 4. QG model results: a) Change in zonal-mean zonal wind for the TSTRAT experiment (see Table 1). b) As in (a) but for the meridional temperature gradient forcing from TSTRAT and the static stability from the control simulation. c) As in (a) but for the meridional temperature gradient forcing from the control simulation and the static stability from TSTRAT. d) As in (a) but for eddy kinetic energy. e) As in (b) but for eddy kinetic energy. f) As in (c) but for eddy kinetic energy.

the PE model, the QG experiments must change both the Held and Suarez (1994) parameter(s) and the static stability, which is taken from the appropriate PE simulation. For TSTRAT (Fig. 4), the QG U change shares the general features of the PE model change: U increases above and on the poleward flank of the jet and there are weaker negative anomalies on the equatorward flank. The amplitude of the QG response is weaker, which is expected given the reduced persistence of the leading U EOF in the QG model (not shown). Changing $\partial\theta/\partial y$ only (i.e. change the Held and Suarez (1994) parameters only), captures most of the U response in the stratosphere and much of the U response in the poleward center of action in the troposphere (Fig. 4b). The $d\Theta/dp$ changes account for a smaller portion of the poleward center of action, but most of the U decreases in the tropics (Fig. 4c). These tropical changes, however, are not present in the PE simulation (Fig. 3a).

TABLE 2. Mechanisms of the response of zonal-mean zonal wind averaged vertically from 1000mb to 700mb (see text).

	$\partial\theta/\partial y$		$d\Theta/dp$		$\partial\theta/\partial y$ and $\overline{\omega}$ on $d\Theta/dp$		$\partial\theta/\partial y$ and $\overline{\omega}_{\text{new}}$ on $d\Theta/dp$	
simulation	slope	corr	slope	corr	slope	corr	slope	corr
TSTRAT	0.74	0.97	0.29	0.93	0.34	0.88	0.26	0.65
$\Delta\theta_z$TROP	0.33	0.62	0.56	0.82	1.11	0.97	1.13	0.97
ΔT_y	0.70	0.85	0.20	0.50	0.95	0.98	0.97	0.98
POLE	0.57	0.93	0.36	0.82	0.99	0.99	0.92	0.99

The relative role of $\partial\theta/\partial y$ and $d\Theta/dp$ on the lower tropospheric average (1000-700mb) U are summarized in Table 2. The "slope" metric gives the regression coefficient, b, that minimizes the difference between the full response and the partial (i.e. either $\partial\theta/\partial y$ or $d\Theta/dp$) response: $\min\,(U_{\text{Full}} - sU_{\text{Partial}})^2$. The minimization is over latitude only, since U is averaged first. An s close to one means the partial response captures the correct amplitude and therefore the corresponding mechanism accounts for the full response. Lower values mean the alternative mechanism is important. The "corr" metric gives the spatial correlation between U_{Full} and U_{Partial}. Low values of this metric can detect cases with "proper" amplitude but "wrong" spatial pattern (i.e. correct amplitude but for the wrong reason). In summary, a partial response does not account for the full response unless *both* the slope and correlation are close to one.

For the TSTRAT experiment, we see that the spatial correlation is large for both temperature gradients ($\partial\theta/\partial y$) and static stability ($d\Theta/dp$) but that the temperature gradients account for about 3/4 of the total response. The relative role of these two processes was also explored in Lorenz and DeWeaver (2007) using a large number of imposed thermal perturbations, however, the conclusions of that analysis were not clear. In a QG model, on the other hand, the temperature gradient and static stability are separate and so a clear quantitative assessment can be made. Finally, when the U response in Fig. 4b and c are added to together we find that the response is very close to Fig. 4a and therefore the linear superposition principle holds (not shown). Superposition also holds for both U and EKE for all remaining experiments discussed below.

The EKE change in the QG model is similar in structure, but the tropospheric anomalies are slightly too weak (Fig. 4d). This is consistent with the weaker tropospheric U response. In the stratosphere the EKE responses to $\partial\theta/\partial y$ and $d\Theta/dp$ are similar in amplitude, in the troposphere,

TABLE 3. Mechanisms of the response of eddy kinetic energy (see text).

	$\partial\theta/\partial y$		$d\Theta/dp$		$\partial\theta/\partial y$ and $\overline{\omega}$ on $d\Theta/dp$		$\partial\theta/\partial y$ and $\overline{\omega}_{\text{new}}$ on $d\Theta/dp$	
simulation	slope	corr	slope	corr	slope	corr	slope	corr
TSTRAT	0.48	0.94	0.43	0.93	0.23	0.71	0.15	0.45
$\Delta\theta_z$TROP	0.41	0.42	0.50	0.51	0.93	0.62	0.98	0.63
ΔT_y	1.05	0.94	-0.05	-0.18	1.15	0.96	1.15	0.96
POLE	0.74	0.93	0.23	0.56	0.99	0.97	0.96	0.97

however, only the $\partial\theta/\partial y$ simulation captures the negative EKE anomalies below and equatorward of the time-mean EKE. To quantify the relative contribution of these two mechanisms to the total response see table 3. The EKE metrics are the same as U (Table 2), except the regressions and correlation are over latitude and pressure instead of only latitude. For TSTRAT, the spatial correlations are similar and $\partial\theta/\partial y$ accounts for slightly more of the total response.

In the next perturbation experiment ($\Delta\theta_z$TROP), the tropical static stability parameter in Held and Suarez (1994) is increased from 10°K to 15°K. The U response in the QG model (Fig. 5a) is broadly similar to the PE model (Fig. 3c), although the amplitude is weaker as before. In this experiment, the response to $d\Theta/dp$ (Fig. 5c) is closer to the full response than $\partial\theta/\partial y$ (Fig. 5b). For example, the former shows a dipole U structure in the mid-latitudes while the latter does not. This is reflected in the slope metrics in table 2: $d\Theta/dp = 0.56$ and $\partial\theta/\partial y = 0.33$. The maximum U amplitude, however, is comparable in the two experiments.

The poleward shift with increase static stability seems consistent with the mechanism in Frierson (2008) and Lu et al. (2008). Even in a QG model, with horizontally uniform static stability, there exists a boundary in the subtropics between baroclinic stable and unstable latitudes. As the static stability increases, the boundary moves poleward, which moves baroclinic instability and the eddy driven jet poleward (Frierson (2008); Lu et al. (2008)). Looking deeper, however, we find that the response is not due to the impact of $d\Theta/dp$ on the waves. In the experiment shown in Fig. 5d, we calculate the heating associated with the control run $\overline{\omega}$ acting on the change in static stability ($\Delta d\Theta/dp$ for $\Delta\theta_z$TROP minus control). This heating tendency is added to the Held and Suarez (1994) physics together with the $\partial\theta/\partial y$ from $\Delta\theta_z$TROP. The static stability, however, remains fixed at the control run values. The U response in this new simulation (Fig. 5d) attains the amplitude of the full simulation in Fig. 5a. This is also documented in Table 2: the spatial correlation and slope

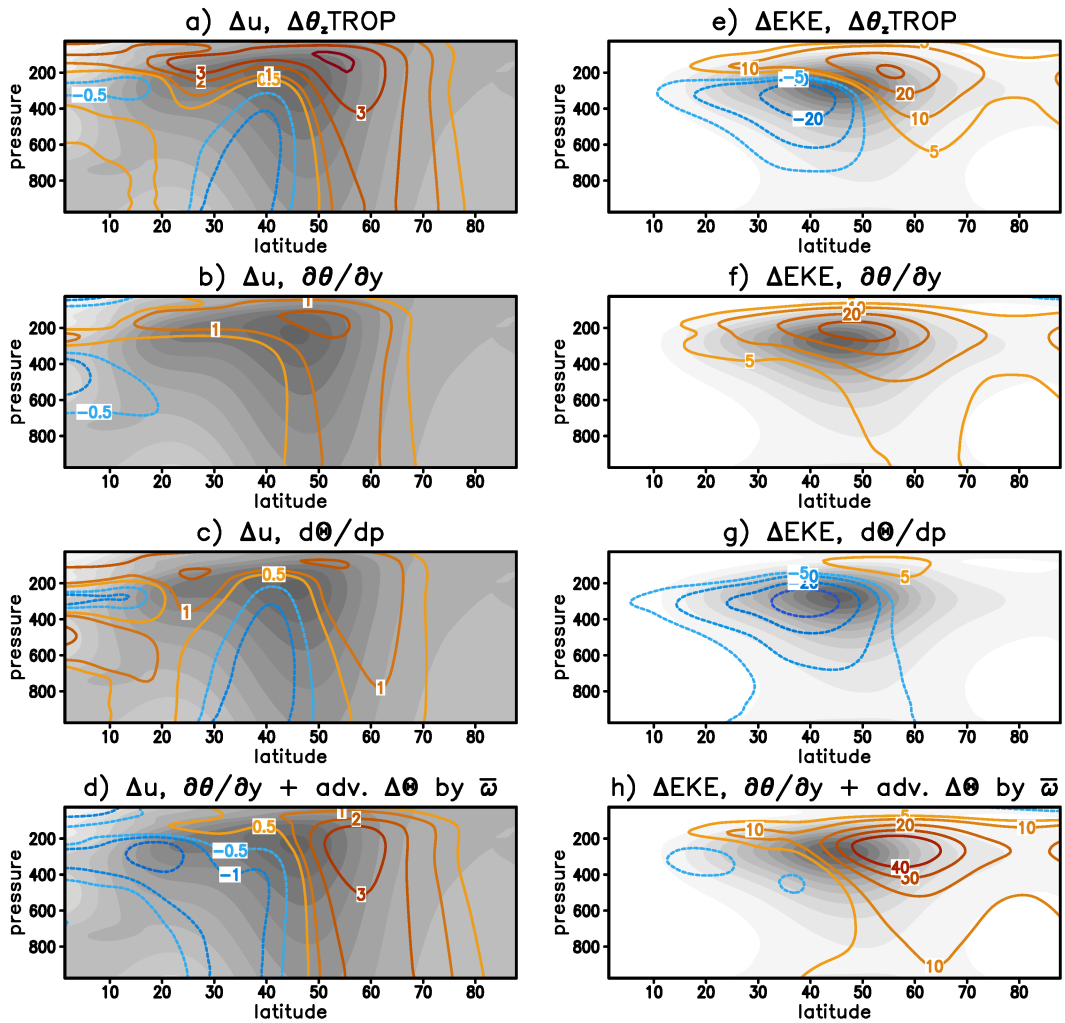

FIG. 5. QG model results: a) Change in zonal-mean zonal wind for the $\Delta\theta_z$TROP experiment (see Table 1). b) As in (a) but for the meridional temperature gradient forcing from $\Delta\theta_z$TROP and the static stability from the control simulation. c) As in (a) but for the meridional temperature gradient forcing from the control simulation and the static stability from $\Delta\theta_z$TROP. d) As in (b) but add heat source equal to zonal-mean ω acting on change in static stability from $\Delta\theta_z$TROP (i.e. vertical advection of Θ by $\overline{\omega}$). e) As in (a) but for eddy kinetic energy. f) As in (b) but for eddy kinetic energy. g) As in (c) but for eddy kinetic energy. h) As in (d) but for eddy kinetic energy.

are 0.97 and 1.11. The mechanism can be seen by looking at the time-mean $\overline{\omega}$ in Fig. 2e, which has downward motion at 28° and upward motion at 57°. When the static stability increases, this leads to heating at 28° and cooling at 57°, which increases the temperature gradient across the jet.

This strengthens the jet and causes a poleward shift consistent with the mechanism in Kidston and Vallis (2012), Lorenz (2014a) and Lorenz (2014b).

One could argue that using $\overline{\omega}$ from control simulation is not the most representative choice because the mean meridional circulation likely weakens as the static stability increases, leading to less vertical advection of Θ. Therefore we repeat the latest experiment but with $\overline{\omega}$ from the new simulation (last column in Table 2). The metrics are almost unchanged, although the details are interesting: $\overline{\omega}$ does weaken slightly but it also shifts slightly poleward to better align with the jet (not shown). Nevertheless, for this particular experiment it is clear that the eddies are not directly involved. Instead, $d\Theta/dp$ shifts the jet poleward via heating from the advection of Θ by the zonal-mean vertical velocity, $\overline{\omega}$.[1]

The EKE response in the QG model (Fig. 5e) is weaker than the PE model (Fig. 3d), which is consistent with the weaker U response. The role of $\partial\theta/\partial y$ versus $d\Theta/dp$ is dramatically different with $\partial\theta/\partial y$ responsible for the increased EKE (Fig. 5f) and $d\Theta/dp$ responsible for the decreased EKE (Fig. 5g). Adding the Θ advection by $\overline{\omega}$ to the $\partial\theta/\partial y$ effect (Fig. 5h) increases the EKE even more except for some weak decreases in the subtropics. Evidently, the decreases in EKE are due to the direct impact of static stability on the waves. This is in contrast to U, where $\partial\theta/\partial y$ and Θ advection by $\overline{\omega}$ explain the total response. We will look at these effects in more detail in section 3d.

In the next perturbation experiment (ΔT_y), the pole-to-equator temperature gradient is increased (Table 1). Like the $\Delta\theta_z$TROP, 1) the $\partial\theta/\partial y$ response is mostly a monopole in the mid-latitudes (Fig. 6b) while the $d\Theta/dp$ response is clearly a dipole (Fig. 6c) and 2) adding the Θ advection by $\overline{\omega}$ to the $\partial\theta/\partial y$ response, we recover the full U response at low levels (Fig. 6d). For EKE, the response is somewhat similar to $\Delta\theta_z$TROP in that the $\partial\theta/\partial y$ response is dominated by EKE increases and the $d\Theta/dp$ response has more prominent EKE decreases (Fig. 6fg). Also, like the $\Delta\theta_z$TROP, adding Θ advection by $\overline{\omega}$ improves the representation of the total response on the poleward flank of the storm track, but not so much on the equatorward flank (Fig. 6h). So, in other words, the EKE decreases in Fig. 6g are caused by the effect of static stability acting directly on the eddies.

[1]Or more precisely, advection by $\overline{\omega}$ strengthens the jet, which leads to a poleward shift via Kidston and Vallis (2012), Lorenz (2014a) and Lorenz (2014b).

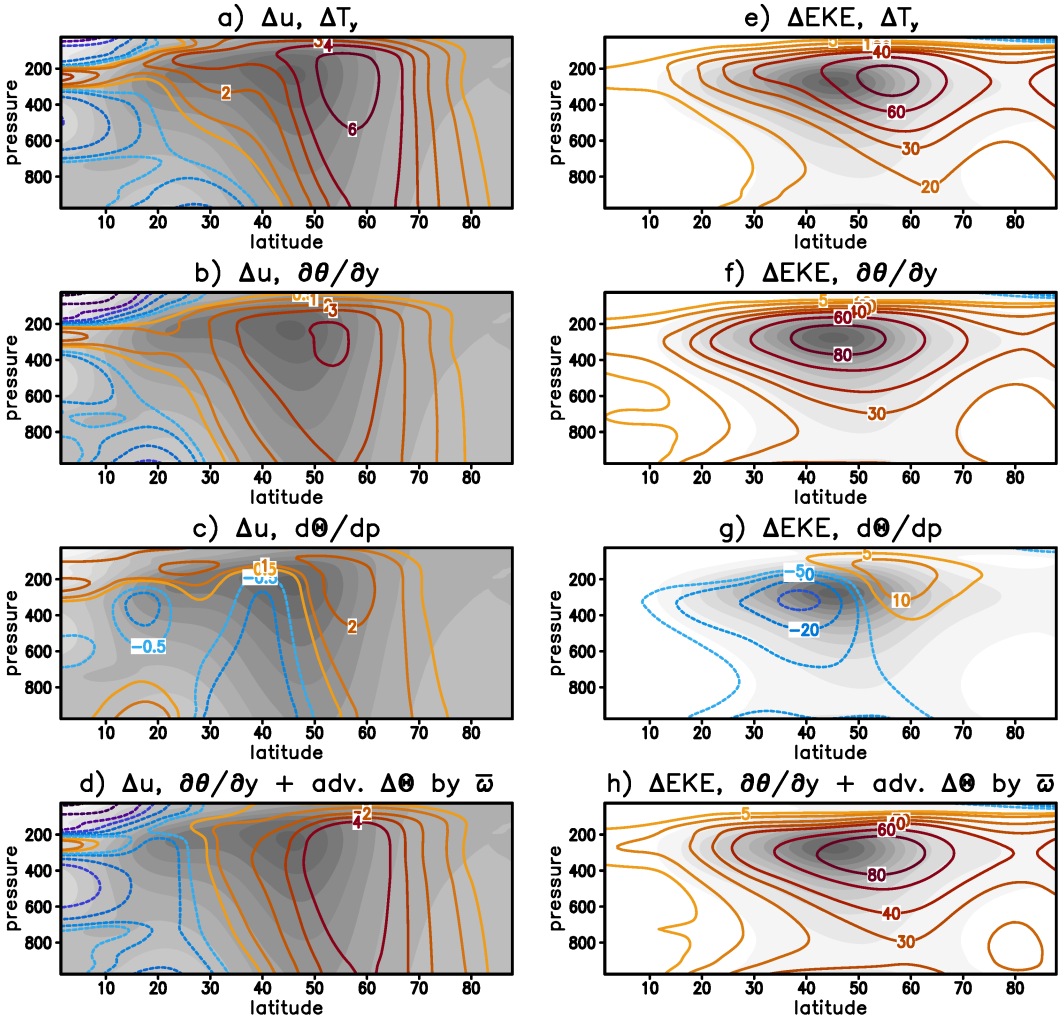

FIG. 6. As in Fig. 5 but for the ΔT_y experiment (see Table 1).

For the final experiment of this section, we show the response to a shallow heating over the polar regions, which is meant to emulate polar amplification under global warming. As found in both simple and comprehensive models (Butler et al. (2010); Peings and Magnusdottir (2014); Deser et al. (2015); Deser et al. (2016); McGraw and Barnes (2016); Wu and Smith (2016); Blackport and Kushner (2017)), the U response is an equatorward shift in the jet in both the QG (Fig. 7a) and PE (Fig. 3d) model. The majority of the U response is due to temperature gradients (Fig. 7b) but the static stability contribution is significant (Fig. 7c). Combining the temperature gradient and the advection of Θ by the $\overline{\omega}$ captures the full response almost completely (Fig. 7d). Therefore, the static stability change does not directly impact the eddies, instead the static stability changes meridional temperature gradients via the zonal-mean meridional circulation.

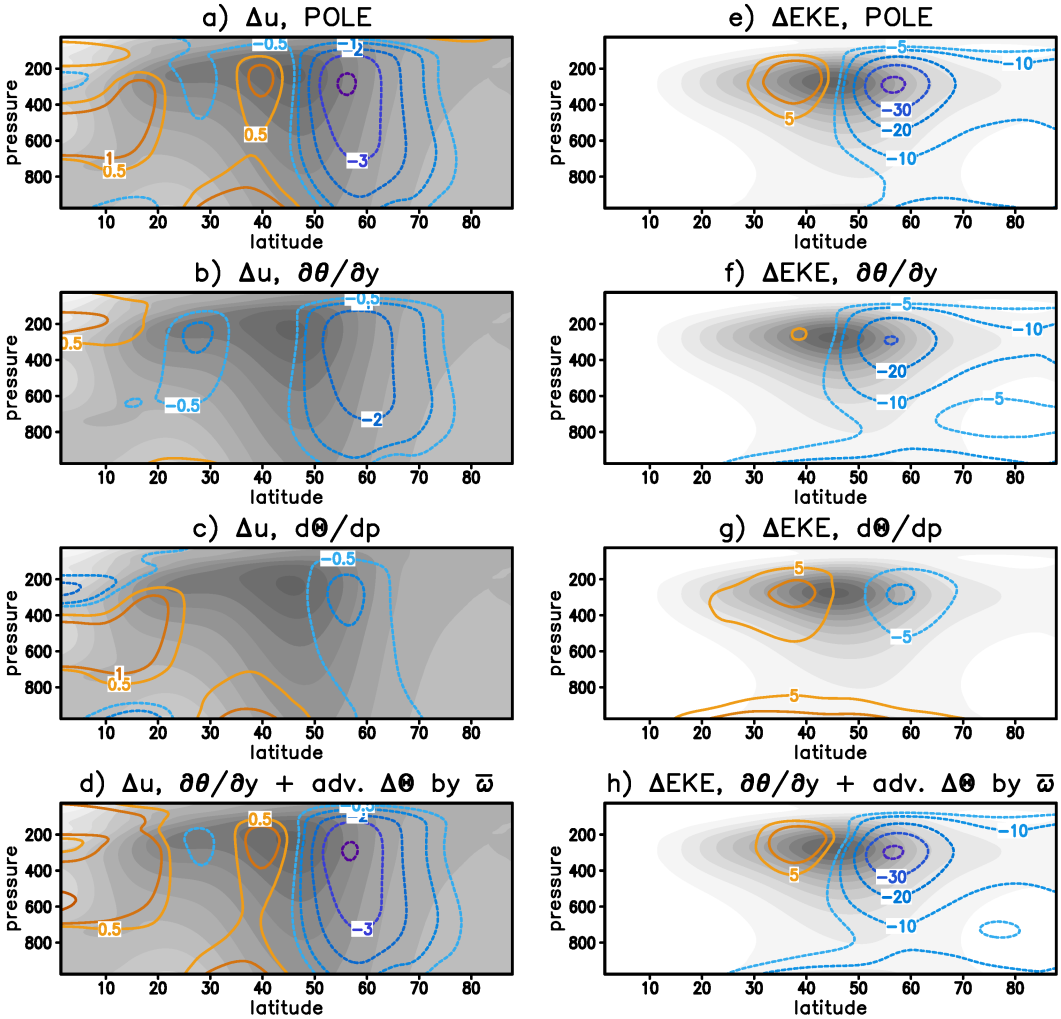

FIG. 7. As in Fig. 5 but for the POLE experiment (see Table 1).

For EKE, the QG model underestimates the response on the equatorward flank of the jet (compare Fig. 7e and Fig. 3h), however, the location of the EKE dipole relative to the mean EKE is reproduced. The changes due to $\partial\theta/\partial y$ are strong and negative on the poleward flank of the mean EKE (Fig. 7f) and the changes due to static stability are a dipole centered on the mean EKE (Fig. 7g). As before, combining the temperature gradient and the advection of Θ by the $\overline{\omega}$ captures the full response almost completely (Fig. 7h). The only component of the static stability response that appears to occur directly via the eddies is the shallow, low-level increase in EKE seen in Fig. 7g. In the full response, however, this is offset by the $\partial\theta/\partial y$ response.

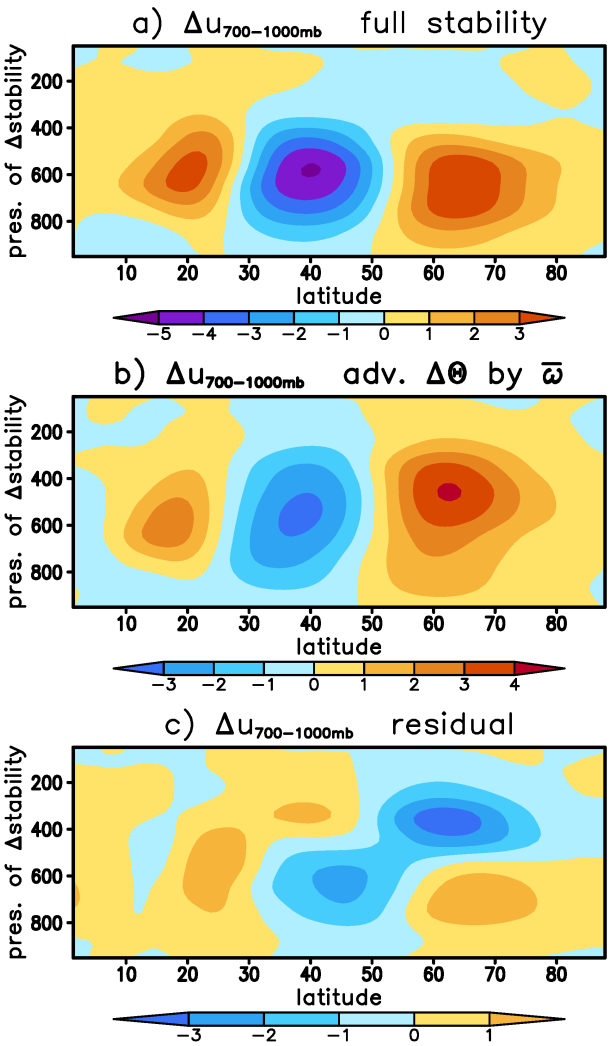

FIG. 8. a) Response of the zonal-mean zonal wind averaged vertically from 700-1000mb (x-axis) as a function of the pressure of the static stability perturbation (y-axis) in the QG model (m/s). b) Same as (a) but the static stability perturbation is replaced by a heat source equal to zonal-mean ω acting on the static stability perturbation (but the actual static stability is kept at control values). c) Residual: response in (a) minus (b).

d. Effect of Static Stability Alone

The QG model provides a unique opportunity to explore the role of static stability alone on the extratropical circulation. Therefore, in this section we explore the circulation response to static stability perturbations that are isolated in the vertical. The static stability perturbations have a Gaussian shape of width 100mb. There are ten experiments, which have perturbation locations ranging from 50mb to 950mb in increments of 100mb (see section 2c).

The U response averaged over the lower troposphere (700-1000mb) as a function of the pressure of the static stability perturbation is shown in Fig. 8a. Any increase in static stability below 400mb causes a poleward shift of the jet (mean jet location $\approx 49°$). Above this level the response is weak. The portion of the total U response caused by the mean control run $\overline{\omega}$ advecting the new background Θ is shown in Fig. 8b. As discussed previously, the latitude of the mean $\overline{\omega}$ (Fig. 2e) implies heating on the equatorward flank of the jet via vertical advection and cooling on the poleward flank. This strengthens the jet, which then causes a poleward shift via the mechanism in Kidston and Vallis (2012), Lorenz (2014a) and Lorenz (2014b). The mean $\overline{\omega}$ advection response explains most of the total U response, although the location of the poleward center of action in Fig. 8b is displaced upward compared to Fig. 8a. The difference between the total and the mean $\overline{\omega}$ effect shows the impact of static stability directly on the eddies (Fig. 8c). This residual response peaks when the static stability perturbation is in the upper troposphere (350-450mb) and causes an equatorward shift in the jet. Nevertheless, this equatorward shift is completely offset by the zonal-mean $\overline{\omega}$ advection. At lower levels, static stability perturbations acting directly on the eddies cause a weak poleward shift of the jet that support the mean $\overline{\omega}$ advection response.

For EKE, horizontally uniform static stability changes primarily affect the vertical structure, therefore we show the global-mean EKE as a function of pressure (Fig. 9). Like Fig. 8, the vertical axis is the pressure of the static stability perturbation, but now the horizontal axis gives the vertical structure of the EKE response. Also, the line $y = x$ is shown for reference. For upper tropospheric static stability perturbations, the EKE decreases from just above the level of the imposed perturbation to the tropopause ($\approx$ 200mb) (Fig. 9a). For lower tropospheric static stability perturbations, the EKE decreases are weaker and are now below the level of the imposed perturbation and extend to the ground. The portion of the response that is a result of the advection of Θ by the mean $\overline{\omega}$ is shown in Fig. 9b. All perturbations lead to a small increase in EKE, which is consistent with the fact that the mean $\overline{\omega}$ is such that increased stability leads to increased meridional temperature gradients via advection. The residual EKE response (Fig. 9c), which shows the direct impact of stability on the eddies, is an enhanced version of Fig. 9a.

The structure of the EKE response may be due to the fact that the critical level for baroclinic instability is in the mid-troposphere[2]. Waves propagating away from the critical level (both upwards

[2]We are not discussing the critical level where waves decay (i.e. subtropical upper troposphere) but the critical level associated with the instability in the mid-troposphere at the latitude of baroclinic wave generation.

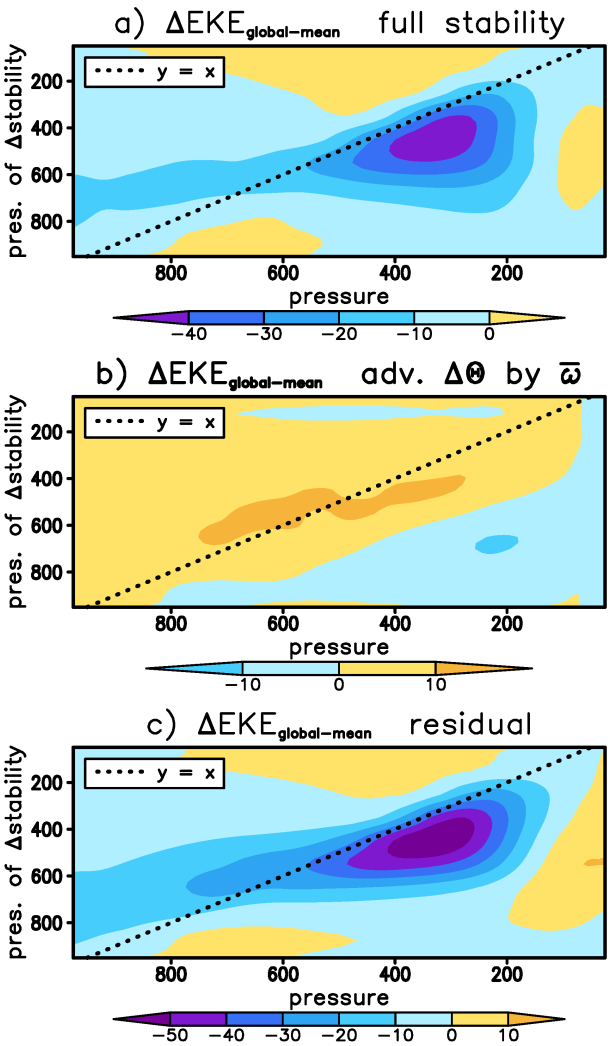

FIG. 9. a) Response of the global-mean EKE in pressure (x-axis) as a function of the pressure of the static stability perturbation (y-axis) in the QG model (m^2/s^2). The line $y = x$ is shown by the thick dotted line. b) Same as (a) but the static stability perturbation is replaced by a heat source equal to zonal-mean ω acting on the static stability perturbation (but the actual static stability is kept at control values). c) Residual: response in (a) minus (b).

and downwards) encounter the increased static stability layer (damping effect) and are therefore weaker after propagating through this layer. This accounts for the fact that the EKE response is above the static stability perturbation in the upper troposphere but below the static stability perturbation in the lower troposphere. For imposed perturbations around 700 to 750mb, which is

a reasonable level for the critical level, the response to static stability is fairly uniform throughout the troposphere.

4. Conclusions

In this paper, a Primitive Equation (PE) model and Quasi-Geostrophic (QG) model are used to understand relative role of meridional temperature gradients and static stability on the response to thermal perturbations in a Held and Suarez (1994) GCM. We take advantage of the fact that the static stability in a QG model is externally prescribed and can therefore be altered independently of the meridional temperature gradient. First, the PE model is integrated to establish the static stability of the control run and each thermal perturbation. Next, the QG model is integrated with the various combinations of control or perturbed Held and Suarez (1994) forcing, and control or perturbed static stability to understand the contribution of each to the full response. In this dry GCM, we focus on the extratropical response only.

Four thermal perturbations to Held and Suarez (1994) were explored in this study: 1) decreasing the stratosphere temperature, which increases the height of the tropopause, 2) increasing the static stability in the tropics, which is also associated with an increased pole-to-equator temperature gradient, 3) increasing the pole-to-equator temperature gradient while keeping the tropopause height at 45° fixed and 4) heating at low-levels over the pole to emulate polar amplification.

For zonal-mean zonal wind (U), static stability is most important for increased static stability in the tropics. For all other experiments, the meridional temperature gradient is more important. Except for increasing the tropopause height, the mechanism for the static stability response, however, does *not* directly involve the eddies, unlike the hypotheses in the literature (Frierson et al. (2006); Frierson (2008); Lu et al. (2008); Kidston et al. (2011)). Instead, the advection of the static stability perturbation by the zonal-mean vertical velocity accounts for almost all of the response to static stability. For increased static stability, advection by the Ferrel cell increases the meridional temperature gradient, which strengthens the jet via thermal wind. The stronger jet then shifts poleward, consistent with the theory of Kidston and Vallis (2012), Lorenz (2014a), and Lorenz (2014b). The opposite response occurs for decreases in static stability. In other words, U is affected by the meridional temperature gradient both directly and indirectly (via static stability). As far as U changes are concerned, the static stability does not affect the eddy field directly. This picture

is quite different from the literature and more work needs to be done to assess the role of this mechanism in moist models and observations.

For Eddy Kinetic Energy (EKE), the direct and indirect meridional temperature gradient explain essentially all of the response for polar amplification. For the other experiments the effect of static stability directly on the eddy field plays a non-negligible role, especially for the increase in tropopause height. However, in all cases, the direct and indirect meridional temperature gradient effects are together the most dominant. We also systematically explored the impact of static stability perturbations localized in the vertical. The EKE response is largest above the perturbation when the perturbation is above 750mb, and vice versa.

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Data availability statement. Please contact the author for the model code and data.

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